

Fall Detection for Seniors

ENG 1102
MT5T3_Phase3_Report
Section: M
Tutorial#: 5
Team#: 3

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1. Design Context & Concept Recap

1.1 Problem Restatement

- Elderly individuals, particularly those aged 65 and older, face significant risks due to falls, especially when living independently. The fear of being left unnoticed after a fall limits mobility and increases anxiety, which further impacts seniors' quality of life. Current fall detection solutions, such as wearable devices or emergency buttons, often fail to address the core issues, including discomfort, inconsistency in use, or the need for manual activation, leaving users vulnerable during emergencies. These devices frequently require users to press a button or interact with the system, which may not be possible when a senior is unconscious.

1.2 Key user insights that guided our prototype

1. **Privacy and Independence:** Seniors need safety without feeling constantly monitored. The devices should provide fall detection without compromising their sense of independence or privacy.
2. **Fear of Delayed Assistance:** The greatest concern for seniors is not the fall itself, but the delay in receiving help. This fear leads to restricted activity and contributes to a sense of isolation.
3. **Comfort and Ease of Use:** Consistent usage is critical. Seniors are more likely to wear a device if it is comfortable and unobtrusive, ensuring it remains functional when needed most.

In Phase 2, we developed three potential concepts: a Smart Fall-Detection Bracelet, a Hip Clip Fall-Detection Device, and a Smart Sole Fall-Detection System. After careful evaluation, the Smart Fall-Detection Bracelet was selected as the most feasible and desirable solution. The bracelet uses an accelerometer to detect falls and a heart-rate sensor to monitor potential health issues, such as fainting, that could lead to a fall. This dual approach ensures seniors receive immediate assistance in the event of a fall or health emergency. The decision to continue with the Smart Fall-Detection Bracelet was based on the results of our decision matrix, where feasibility (38%) and desirability (26%) were prioritized. The bracelet performed well in both areas, with its use of readily available technologies such as motion sensors and heart-rate monitoring, making it technically achievable. Its wearable, non-intrusive design also addresses users' need for comfort and discreet safety, critical for encouraging consistent use. Therefore, the Smart Fall-Detection Bracelet is the optimal choice to meet the functional and emotional needs identified through user research.

2. Prototype Development

Main Question: Can an accelerometer reliably detect a fall and filter it from ordinary movements such as waving or clapping?

Prototype Type:

The prototype was built as a proof of concept because the greatest uncertainty in the design was whether the sensing and detection system would actually work reliably. Instead of jumping straight into a finished wearable, the focus was on testing that core functionality first. This made it easier to quickly validate the algorithm, keep costs low, and adjust the detection settings without limiting a finalized design.

Prototype Description

The prototype is built around an ESP32-S3 microcontroller board, which houses a built-in 6-axis IMU (3-axis linear accelerometer and 3-axis gyroscope). The system was monitored in real time through the Arduino IDE serial plotter and serial monitor via USB connection to a laptop. The detection logic was implemented as a Finite State Machine (FSM), which governs all state transitions. The system provides a range of features, from idle monitoring and fall detection to alerts and SOS dispatch.

One of the key design decisions was to use a single combined acceleration value instead of looking at each axis separately. Early testing showed that when a fall happens, the force spreads across multiple axes at the same time, so no single axis consistently crosses the threshold on its own. To address the problem, the total magnitude of the acceleration (both linear and angular) was calculated by combining all three axes into one value. This made the system much more consistent at detecting falls, no matter the direction (forward, backward, or sideways).

The FSM detection logic required two simultaneous conditions to be met before classifying a motion event as a fall:

1. Linear acceleration magnitude: exceeding 4 m/s^2 , indicating a high-impact motion event.
2. Angular acceleration magnitude: exceeding 100 m/s^2 , confirming a significant rotational shift in body posture rather than a brief routine gesture.

These threshold values were not assumed; they were determined experimentally through controlled drop trials and real fall simulations.

Upon detection, the FSM transitions to an alert state: the OLED display activates, showing “DO YOU NEED HELP?” alongside a 20-second countdown. If the user responds by pressing the input button, the system resets to monitoring mode. If no response is received within 20 seconds, the FSM transitions to its final state, and the OLED displays “HELP IS ON THE WAY,” simulating an automatic emergency alert to a caregiver. This fail-safe directly addresses the core user risk identified in research: an unconscious senior cannot manually call for help.

A heart rate sensor is planned for integration in the next development stage. This will allow the device to detect health events such as fainting that may precede a fall, enabling the bracelet to alert caregivers proactively rather than only reactively.

Key Components and How They Interact

Component	Role	Interaction
ESP32-S3 Microcontroller	Processing unit	Runs the FSM, reads IMU data, controls all outputs
Built-in 6-Axis IMU	Motion sensing	3-axis linear and 3-axis angular acceleration data to the ESP32-S3 at high frequency
FSM (Firmware Logic)	Fall detection engine	Evaluates acceleration against thresholds and handles all state transitions
2.06" OLED Display	User alert interface	Displays “DO YOU NEED HELP?”, 20-second countdown, and “HELP IS ON THE WAY” upon fall detection
Input Button	User response	Cancels the alert if pressed within the 20-second window
Laptop (USB)	Power and monitoring	Powers the ESP32-S3; displays real-time sensor data

Materials used

- ESP32-S3 microcontroller board (with integrated 6-axis IMU)
- 2.06" OLED Display
- Momentary push button
- Jumper wires and solderless breadboard
- USB cable
- Laptop running Arduino IDE



- Photo 1 (top left) – Idle State: The prototype displays a digital clock face (12:00). A small green indicator dot confirms the system is active and continuously reading IMU data. This is the default state the device remains in until a fall event is detected.
- Photo 2 (top center) – Fall Detected/User Help Check State: Upon the IMU exceeding both the linear acceleration threshold (4 m/s^2) and the angular acceleration threshold (100 m/s^2), the FSM transitions out of idle. The display immediately switches to a red screen reading "FALL DETECTED!" and prompts, "Do you need medical help?" with clearly labeled YES (green) and NO (red) buttons. This state gives the user 20 seconds to respond.
- Photo 3 (top right) — SOS Sent State (Yes selected): If the user confirms that they need help, the FSM transitions to the SOS state. The display shows "HELP IS ON ITS WAY" alongside an orange medical cross symbol, simulating an automatic emergency alert being dispatched to a caregiver.
- Photo 4 (bottom left) — Reset State (No selected): If the user indicates they are okay, the FSM displays "Glad you're OK!" alongside a smiley face icon before returning the device to the Idle state. This prevents unnecessary emergency alerts while still acknowledging the detected event.

3. Testing & Results

The tests for the medical alert bracelet focus on validating the IMU (inertial measurement unit) sensor performance (accuracy) and the FSM (finite state machine). The IMU was tested by running multiple trials under controlled conditions to determine whether it could accurately detect motion on all six axes of inertia. Then, the reading of the inertias for each axis tested under constant conditions (accelerations) was reviewed to determine and eliminate noise (using a moving average algorithm) from the readings, reducing future false positives in the fall detection algorithm. Next, the FSM (finite state machine) that determined if the user had fallen was tested. The FSM was tested using the determined values of linear and angular acceleration determined experimentally (by falling and seeing the readings of the maximum linear and angular acceleration). The method was used to determine if the FSM was transitioning or idling in the correct state under both normal and fall event conditions. Multiple trials were conducted, where 96 out of 100 successful detections/state transitions occurred. The exact tests are shown in 3.1.

3.1 Test Plans:

Test Plan Field (1)	Testing the IMU (inertial measuring unit) on the ESP32-S3 board
Feature being tested	IMU reads acceleration in 6 axes (3 linear, 3 rotational)
Testing method	Connect the ESP32-S3 to microcontroller logic and attempt to read values from the sensor every millisecond. Move the IMU along each axis to determine if the IMU is reading acceleration for every axis. Display the results via a live graph to the Arduino IDE.
Number of trials	Ten trials will be conducted by dropping the sensor from the same height. Each trial should detect motion along each axis. Help determine anomalies early to make future prototyping more manageable and less time-consuming.
Success criteria	The IMU must be able to read any changes in acceleration at a minimum of 3 m/s ² linear and 50 m/s ² angular to detect a rapid change in acceleration that indicates a fall.
Potential risks / mitigation.	Risk: Sensor noise causes readings in other axes or the axis in which we are testing. This change would lead to a reduction in accuracy in the fall detection mechanism as a whole. Mitigation: Applying data filtering techniques like a moving average noise mitigation algorithm can help reduce false detections of acceleration changes.

Test Plan Field (2)	Testing the FSM (Finite State Machine) for accurate fall detection.
Feature being tested	Correct state transitions/behavior under fall conditions.
Testing method	Provide controlled inputs to the FSM to verify that the states change only under the specified conditions. Use a SerialPort monitor in the Arduino IDE or the prototype's display to determine correct state transitions under normal use conditions.
Number of trials	3 trials, each with 10, 20, and 100 drop/fall tests, respectively, are to be conducted. Each trial should determine if all possible transition states are occurring under their specified conditions. The goal is to prove that the states occur only under correct events, consistently.
Success criteria	The FSM transitions to the next state correctly under valid IMU input data. No unexpected states occurred within 95% accuracy.
Potential risks / mitigation.	Risk: Logic errors within transition states and noise propagation throughout the system can cause unpredictable states. Mitigation: Using input filters, performing unit testing, logging events, and defaulting states and error states should help mitigate the propagation of errors throughout the system.

Test Plan Field (3)	Testing the comfort and usability.
Feature being tested	Comfort for users in wearing and operating the device.
Testing method	Have product testers (our elderly neighbors) wear and use the device.
Number of trials	10 trials conducted. These trials were to determine the user's potential needs for future prototyping. It will also help address current issues that can be spot-fixed before the following iterations.
Success criteria	The user gave over a 7.5/10 rating in comfort and 8/10 for usability. This was done using anonymous polling.
Potential risks / mitigation.	Risk: Users have a difficult time using the device's UI, or they use the device in conditions where it can't operate (underwater). Mitigation: Demonstrate how to use the device to the user before they test it. Explain the restrictions for device use in specific conditions clearly to the user.

3.2 Testing Results & Analysis

Quantitative & Functional Testing

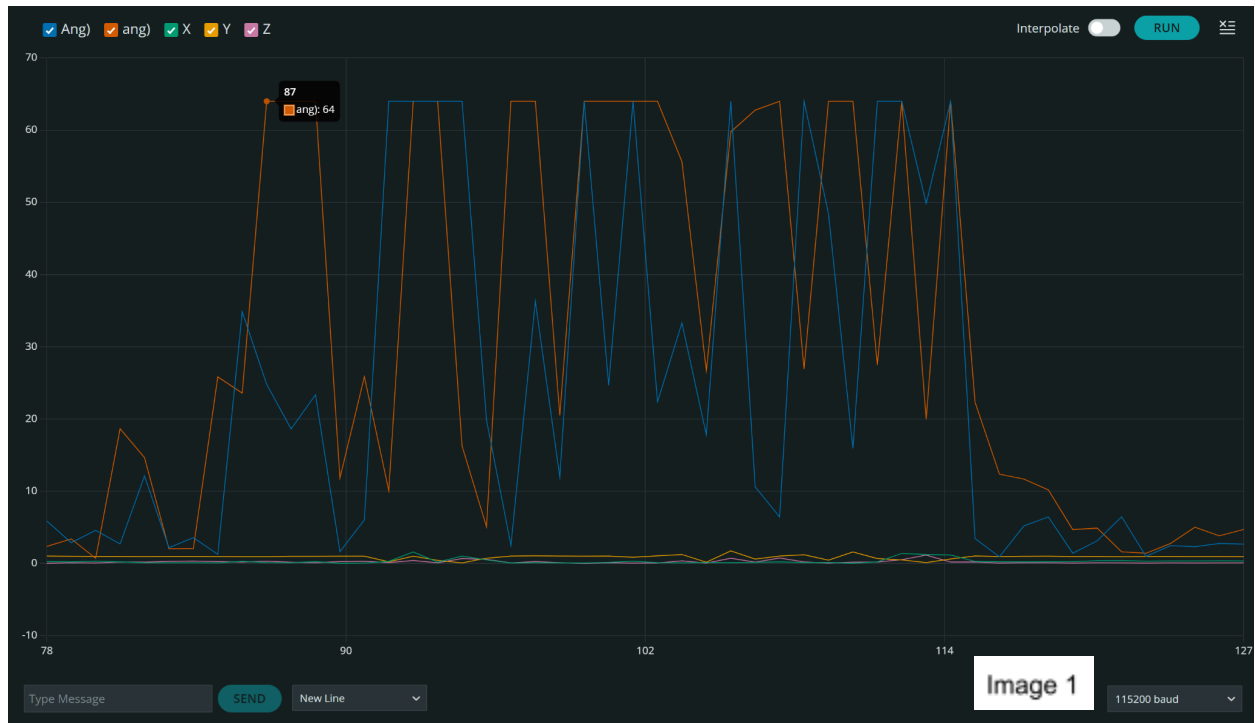
IMU TEST (Quantitative): To test that the IMU detects acceleration consistently and accurately, we perform 10 controlled drop trials to ensure that each axis can detect accelerations (as displayed in the picture below). From all 10 trials, we determined that the angular acceleration in most falls could range from 20 to 70 m/s^2 and the linear accelerations from 0 to 6 m/s^2 . In testing for system noise, we first used a flat plane (a table) and moved the IMU in a straight line along an imaginary y-axis, then an x-axis, and finally diagonally to determine if acceleration was detected only in the y, x, and then y and x directions. We conducted a similar test with the angular axes by tipping the IMU over along the rotational axis in question and measuring any possible deviations in the other rotational axes. The result was that the

The system was surprisingly quiet. We still ended up using a moving average filtration to smooth out random hardware spikes that occur in almost all microcontroller acceleration systems.

Unexpected Results: From the x,y plane acceleration trial, we realized that the total acceleration would be split into an x component and a y component, which could cause our FSM to not activate the next state if the accelerations in each axis were not high enough, even if the total acceleration were around the threshold.

Fix: Therefore, we implemented a unified angular and linear acceleration magnitude to determine if a fall occurred due to the fact that a falling vector could have force and moment components in each axis. The information is displayed in the second image. The formula used to determine this value was based on the magnitudes of the linear components and the angular components.

Seeing that the sensor was able to detect within our researched and tested success criteria outlined in Test 1, with minimal noise in the system, this demonstrated a successful test, allowing us to move further on with our prototyping.



FSM TEST (Quantitative): Now that our IMU was working and able to input accurate unified (magnitude) linear and angular acceleration into our finite state machine, it was time to test it. We started with trial one, where we dropped the IMU 10 times to determine an initial range of the unified linear and angular fall accelerations. Taking the average of all the drop values, we

determine a base for our change of state to detect a fall in our FSM. For the second trial, we tested our new FSM using the determined values from the first trial. The key difference between these two trials was that now the trial was testing a real fall using our physical watch prototype instead of just dropping the IMU. For this trial, we each took turns falling forwards, backwards, and side-to-side to further tune our state-change logic in determining a fall.

Results of trials 1 & 2: The resulting trials lead us to determine linear and angular acceleration thresholds for our third and final FSM testing in detecting a fall. The thresholds that we determined from the two trials were that a linear acceleration of 4 m/s^2 and angular acceleration of 100 m/s^2 were a suitable choice when wanting a high activation rate of state change and accuracy.

Trial 3 & unexpected results: Taking what we learned from the first two trials and implementing it, we then ran a final trail (trial 3) to test the accuracy of our thresholds under normal conditions and fall conditions. Out of 100 tests, we only observed 2 false positives under normal use conditions and 2 undetected fall states. Therefore, our FSM was observed to have a 96 percent accuracy rate, which is above our success criteria of 95 percent accuracy.

Fix: We did not observe any further false positive activations under normal conditions. This made us believe they were potentially just a fluke. No changes were made, seeing that our criteria of success were already high (95% accuracy) and met. For future iterations, adding additional filtering and testing for even higher accuracy is recommended.

User Feedback & Usability Testing (Test 3)

Summary of User Results	
Metric	Result
Number of Elderly (65+) Participants	10
Average completion time	10 mins
Ease-of-use rating (1-10)	8.2 / 10
Comfortability rating (1-10)	9.1 / 10

Qualitative themes:

- A majority of your users found it quite comfortable to wear. A few stated that “It was as comfortable as their watch” or that it “didn’t feel like they were wearing a medical device.”
- About half of the testers understood how to use the device, while the others needed to be shown a few times until they understood how to operate it.
- Users suggested that the device should have more ways to indicate when an emergency event is triggered. (sound system, wrist vibration).

User Testimonials:

- “It feels nice to wear, but I wasn't always sure that I was using it correctly.”
- “It's very light and slick. My only complaint is that I want to be able to swim with it.”
- “Once it was explained to me, it was easier to operate. I hope I don’t forget.”

Test 3 comparison test results: The goal of test 3 was to determine if the final prototype (P3) was good enough for the user to actually use. After testing the device on 10 users and gathering their feedback and ratings, we determined that it was technically successful because their ratings exceeded our threshold for a successful test. One thing we noticed was that many future improvements can still be made, which we didn’t know needed fixing until we heard the testers' testimonials, such as more interactivity like sound, vibrations, etc. The ratings from the testers for ease of use and comfort were 8.2 and 9.1 out of ten, respectively. The score was above our success criteria of 7.5 and 8 out of ten. Therefore, it passes our requirements for the prototype.

Unexpected User Behavior:

- Users frequently looked to verify that the device was on and functioning.
- Some users forget parts of the instructions, which suggests that we can rely on memory alone for them to be able to operate the device.

Implication: These behaviors suggest that the design, assuming that users can rely on their memory, could eventually become a problem for this current iteration of the device.

Interpretation of Results:

- Which aspects of your design did the evidence support?
 - The device met the comfort (8.2/10) and usability requirements (9.1/10) after guidance from designers (us). This shows the design is both physically and functionally usable. Future iterations should, however, solve users' new needs.
- Which assumptions turned out to be incorrect or incomplete?
 - This assumes that the user will remember how to use the device and understand its feedback. This was proven by the user's uncertainty during use.
- What did your test reveal that your prototype planning had not anticipated?

- Users relied heavily on continuous visual feedback and also forgot periodically how to use the device. They also preferred real-time guidance over video explanations.
- How do the usability findings connect to your iteration log? Did user feedback directly drive a prototype change?
 - Future iterations should focus on clearer feedback to the user (audio, vibration, etc.) and also create more autonomy for the device, as users sometimes tend to forget how to operate it.

3.3 Iteration Log

Iteration	Finding / Failure	Change Made	Rationale	Outcome
P1 → P2	The IMU axes and FSM would not detect multiple types of falls because the total linear and angular acceleration is broken into components that are individually too small to detect a fall and trigger the next state.	Implement unified linear and angular acceleration values by taking the magnitude of all the linear axes and the magnitude of all the angular axes.	Eliminates a majority of undetected falls due to component vectors not passing their individual thresholds. Also simplifies the logic for handling states.	It increases accuracy by about 15 percent. This event occurred between test 2's 2nd and 3rd trials.
P2 → P3	Added state machine for automatic EMS call determination. The current prototype could not activate if the user was unconscious after a fall, preventing the emergency state from activating.	If a fall is detected and the user does not respond, the FSM assumes that the person has had a medical emergency after 30 seconds of no response and contacts EMS on their behalf.	20% of elderly people lose consciousness after a fall. Therefore, the system must be able to handle this event.	Increase in potential survival rates for elderly individuals after suffering a medical emergency due to a fall.

3.4 Critical Reflection

- What was the most important thing you learned from testing that you could not have known from Phase 2 alone?
 - The most important thing we learned that we couldn't have known from phase two was that a prototype, in theory, can look perfect under very particular assumptions made by the development team. We ran into this issue when testing our prototype with the user, believing they would be able to remember a few instructions to operate it. We came to realize that this was not always the case because some users have issues with remembering things in general. For future designing, we should ask much more specific questions like "Do you experience problems with your memory?" So better tailoring of initial questions will reduce false assumptions, which will have a more positive impact on design ideation/choice and prototyping phases.
- If you continued to a higher-fidelity prototype, what would be the single most important change to make?
 - Use user feedback to add more indicators, like vibration and sound, to make the device more intuitive.
- What aspects of usability or user experience remain unresolved, and what additional testing would you conduct if more time or resources were available?
 - Testing users' lack of confidence and forgetfulness will help us develop more effective FSM feedback and emergency activation methods that require user intervention only if desired.
- What would be improved if more time/resources were available?
 - Improvements include increased fall detection accuracy, true waterproofing (as requested by users), further refinements to the user interface, and testing with larger groups of users to enhance reliability and usability in the future.

4. Value Proposition Canvas

4.1 Customer Profile

Customer Jobs

The primary users are elderly individuals who want to maintain independence while ensuring their safety. Their main goal is to perform daily activities confidently without fearing that someone will leave them unattended in case of a fall.

Pains

Users strongly fear falling and being left without immediate assistance. Many existing devices are uncomfortable, difficult to use, or unreliable due to false alarms. Delayed emergency response and complex interfaces further reduce user trust and willingness to adopt these solutions.

Gains

Users seek faster emergency assistance, improved safety, and greater independence. They prefer comfortable devices that are simple to use and require little to no interaction. Reducing caregiver anxiety is also an important benefit.

4.2 Value Map

Products & Services

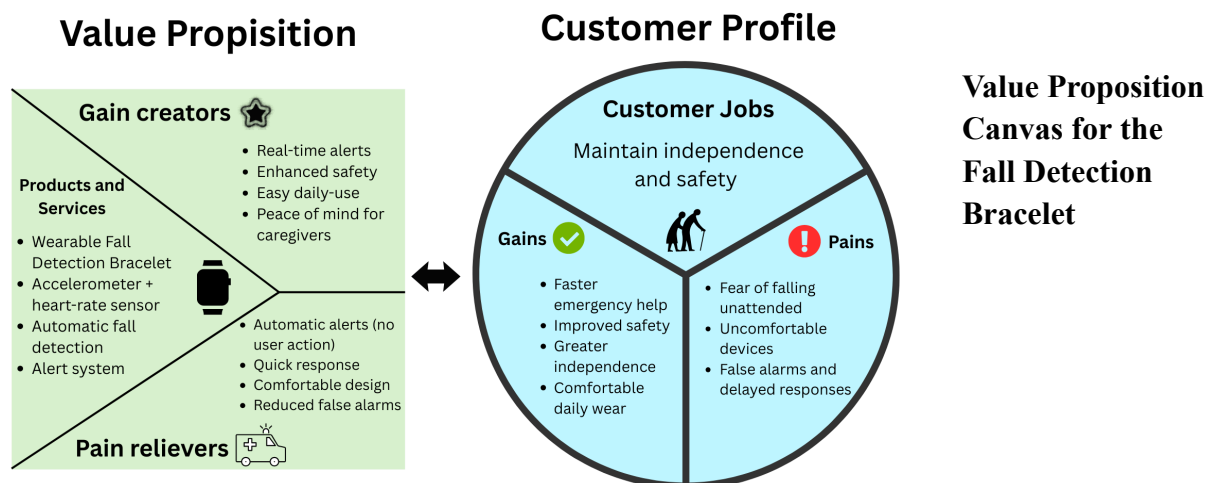
The proposed solution is a wearable bracelet that uses an accelerometer and heart-rate sensor to automatically detect falls and trigger alerts without requiring user input.

Pain Relievers

The bracelet eliminates the need for manual activation by automatically detecting falls, which reduces delays in emergency response. Its comfortable and familiar design encourages consistent use, while improved detection helps minimize false alarms.

Gain Creators

The device provides real-time alerts, improving response time and user safety. Its wearable form factor increases comfort and usability, allowing users to maintain independence while feeling secure. It also offers caregivers peace of mind.



5. Implementation consideration

5.1 Safety

- **The Challenge:** As a team, we analyzed how the device sits on a senior's wrist and identified three main hazards. First, wearing a device 24/7 in a damp environment can cause skin irritation or rashes if the material isn't hygienic. Second, we worried about a "silent failure" due to user distress: what if the user falls and is knocked unconscious, making them unable to interact with the OLED screen? Finally, there is the risk of technical failure; because the current prototype is not waterproof, exposure to high humidity or water in the bathroom could cause the device to short-circuit and stop working entirely without the user realizing it.

- **Our Mitigation Strategy:**

1. To keep the user safe, we chose rubber silicone straps. Because silicone is non-porous and water-resistant, it won't trap bacteria, helping to prevent skin infections. However, we must clarify that the current prototype is not waterproof; it can handle minor splashes but cannot be worn in the shower.

To ensure 24/7 protection and hardware reliability, our next implementation goal is to upgrade the casing to a fully waterproof (e.g., IP68-rated) design.

2. For the emergency risk, we agreed on a 20-second fail-safe. If the "DO YOU NEED HELP?" prompt isn't answered, the device assumes the user is unconscious and automatically sends the SOS.

5.2. Reliability

The Challenge: For reliability, our team asked a tough question: *"Is the system going to detect a fall every time the user just claps or waves their arms?"* If the device constantly sends a "false alarm," the user will become frustrated and stop wearing it. The challenge is making the 6-axis (including the rotation) accelerometer smart enough to tell the difference between a celebration and a dangerous fall.

- **Our Reliability Strategy:** To ensure the system stays dependable, we programmed the 6-axis accelerometer to look for a dual trigger. The software doesn't just look for a sudden "jerk" (linear acceleration); it also checks for a significant change in body angle (angular acceleration).
 - Clapping or Waving: These movements might be quick, but the wrist usually stays in a similar position relative to the body. The sensor recognizes such activity as "Normal Activity" and ignores it.

- A Real Fall: This involves a high-impact spike followed by a total shift in orientation.

By requiring both these events before the OLED screen activates, we have ensured the device's reliability for daily use, preventing false alarms during routine tasks.

5.3 Manufacturing & Assembly

As a team, we realized that our current prototype has a big limitation: it only works when it is plugged into a computer or Arduino. In the real world, a senior cannot walk around the house or take a shower with an Arduino and several loose wires. This is the main "problem" we had to solve when contemplating a finished product.

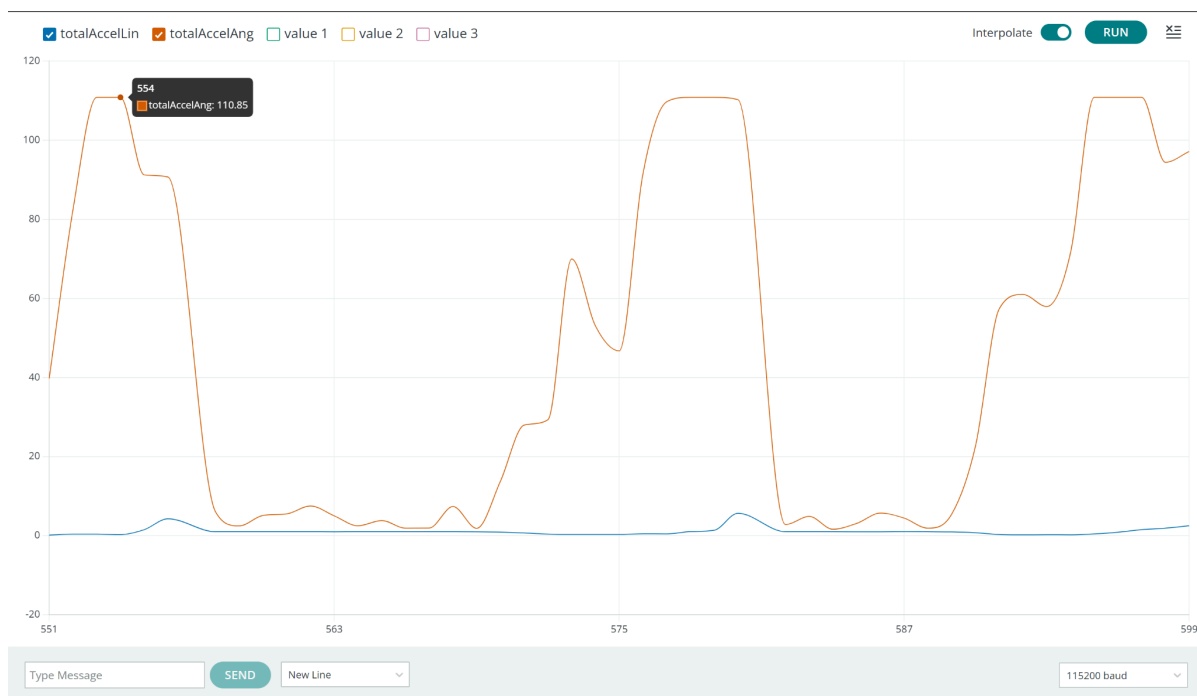
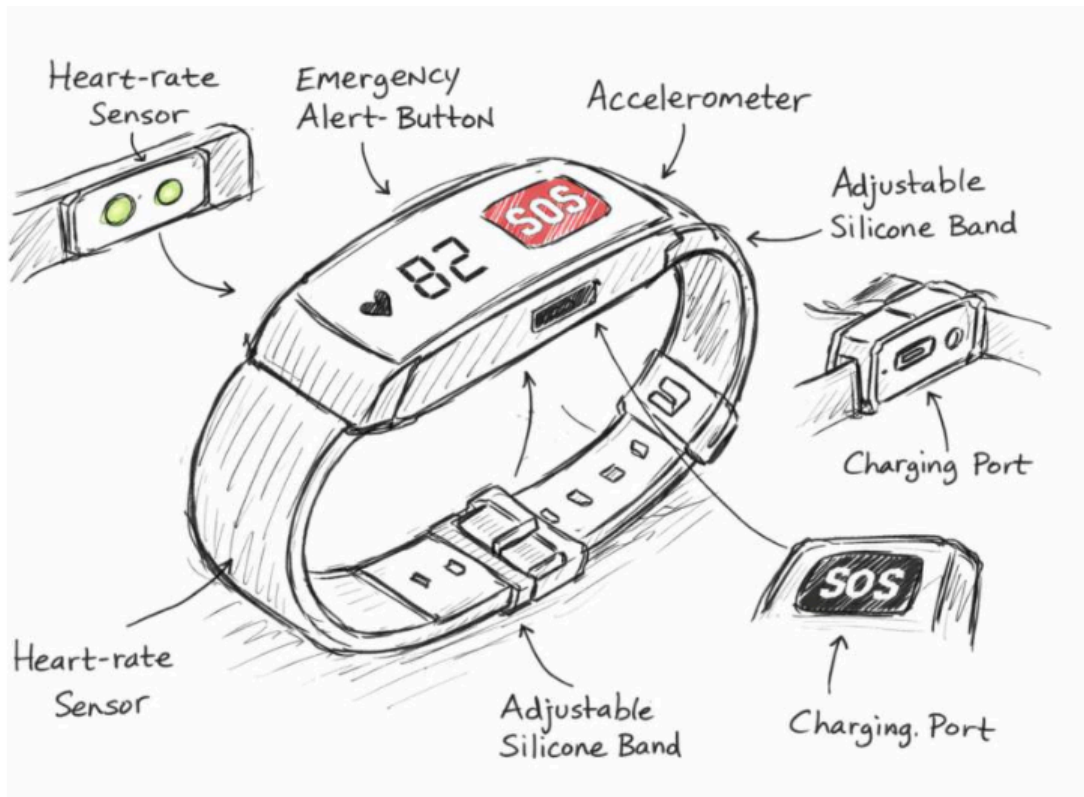
- **The Challenge:** How do we fit the accelerometer, heart rate monitor, buzzer, and LED lights into a small bracelet without needing a computer to run the code or provide power?
- **Our Feasible Solution:** To make this work in real life, we would move away from the big Arduino board and use a tiny, custom-made circuit board (PCB).
 - No More Computers: We would upload our code directly onto a small computer chip on that board, so it runs by itself.
 - Battery Power: Instead of a USB cable, we would use a small rechargeable battery (like the ones in a smartphone) tucked inside the case.
 - Simplified Assembly: We chose injection molding for the plastic body because it allows us to snap the small board and battery into place perfectly. This keeps the device lightweight and comfortable for the user's wrist while making sure all the parts stay connected and waterproof.

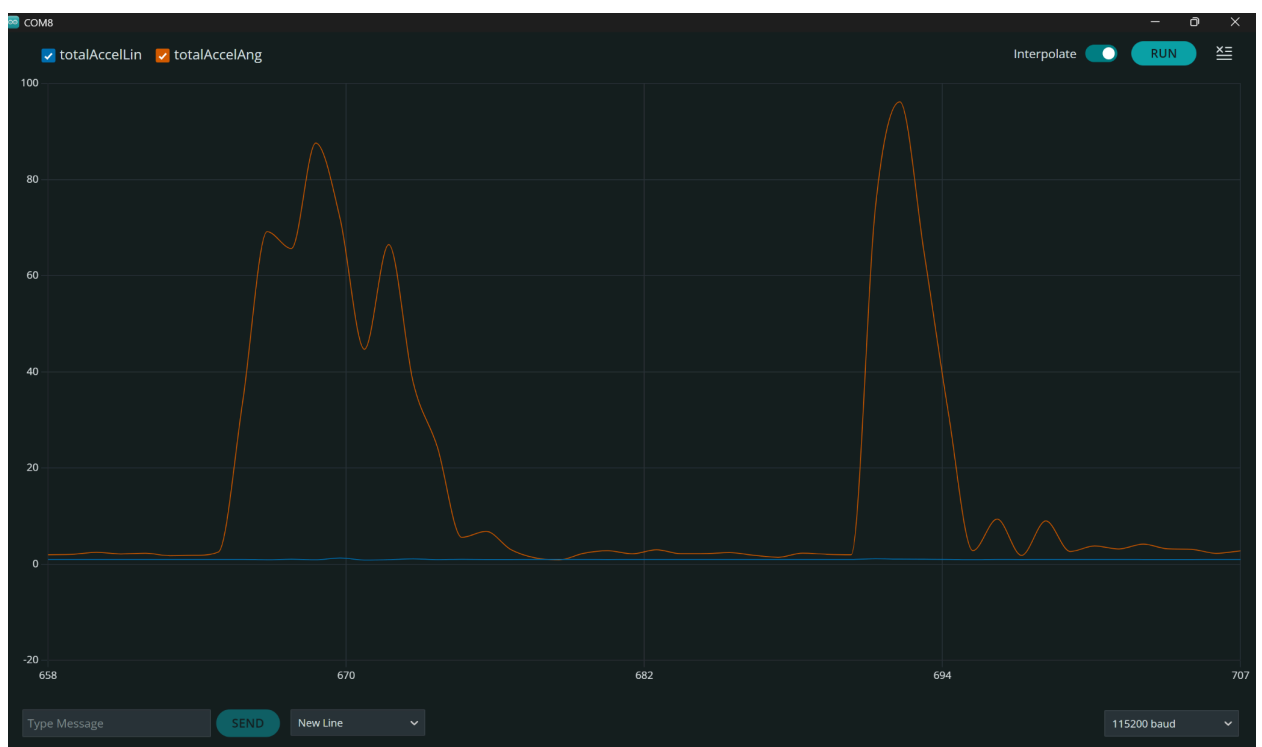
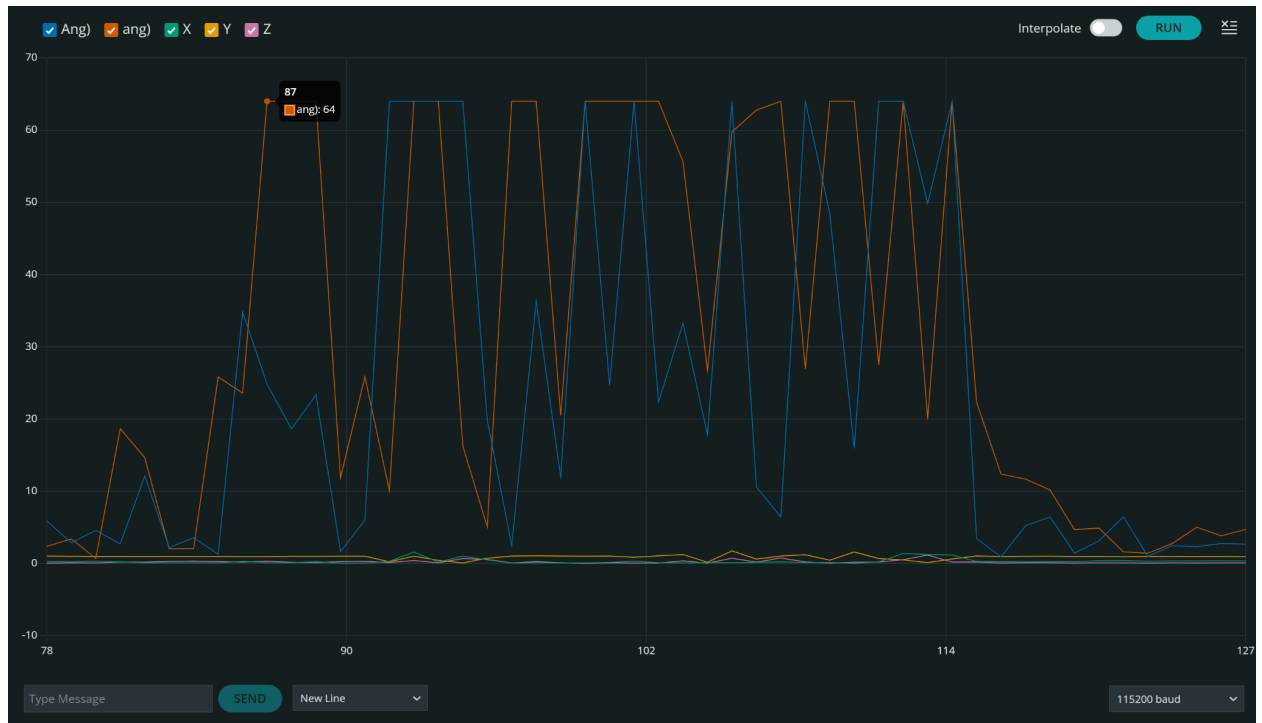
6. Attribution Table

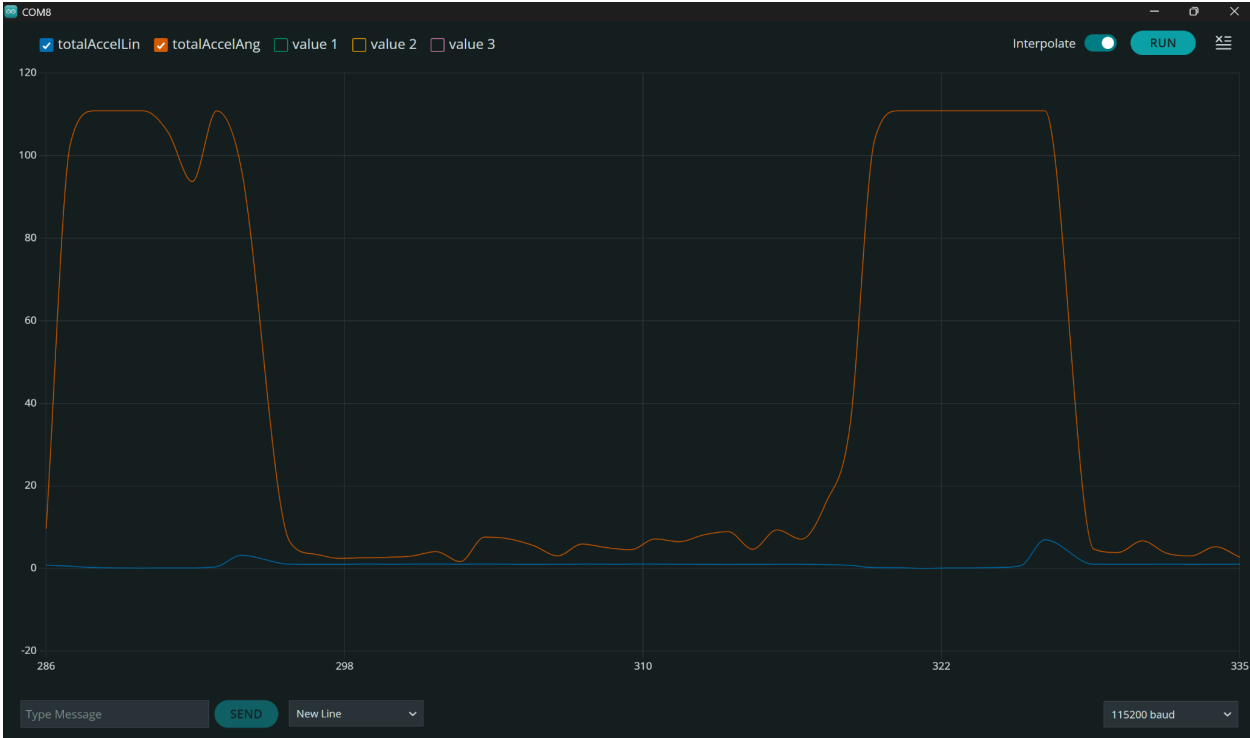
Student	Type of Task (Design / Research / Teamwork / Project Management / Communication)	Contribution	Approx. Time / Effort (Low/Medium/High)
Noah	Design, Research, Communication, Teamwork	- Completed Section 3 of the Phase 3 report. - Did a majority of the hardware implementation of our prototyping and testing. - Created an activation graph for the poster. - Conducted research on hardware to use for the prototyping phase. - Created the FSM for the ESP32 board (the logic of the prototype) - I attended team meetings to determine our team's progress and my individual progress.	High
Riana	Design, Project Management, Teamwork, Communication	- Completed section 4 of Phase 3 Report, aligning user needs with the proposed solution - Set internal deadlines and ensure all team members completed their assigned tasks on time by tracking progress - Developed a draft version of the prototype code for the final bracelet concept - Created a draft and edited the final poster - I attended team meetings to determine our team's progress and my individual progress. - Ensured each section met the rubric requirements for writing quality and organization.	High
Reham	Design, Project Management, Teamwork, Communication	- Completed section 1 of Phase 3 Report - Helped with draft version of prototype code for final bracelet concept - Collaborated on the visual layout and content of the presentation poster. - I attended team meetings to determine our team's progress and my individual progress.	High
Emelyne	Research, Communication, Teamwork	- Conducted research about the best straps for our smart bracelet fall detector.	High

		- Collaborated on the visual layout and content of the presentation poster. - Completed section 5 (implementation consideration). - I attended team meetings to determine our team's progress and my individual progress.	
Sarvesh	Design, Research, Communication and Teamwork	- Wrote and finished section 2: Prototype Development of Phase 3 report - Documented the proof of concept prototype design, ie, component roles/interactions - I attended team meetings to determine our team's progress and my individual progress. - Researched and outlined the ESP32-S3 and 6-axis IMU technical specifications - Sourced and annotated visual documentation, including FSM state transition photographs - Ensured each section met the rubric requirements for writing quality	High

7. Appendices







Moving average formulas for filtering.

$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$	
$S = \sum_{i=1}^m r_i^2 = \sum_{i=1}^n (y_i - (a + bx_i))^2$	
$a = \frac{\bar{y}(\sum_{i=1}^n x_i^2) - \bar{x} \sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$	$\frac{\partial S}{\partial a} = -2 \sum_{i=1}^n (y_i - (a + bx_i))$
$b = \frac{(\sum_{i=1}^n x_i y_i) - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$	$\frac{\partial S}{\partial b} = -2 \sum_{i=1}^n (y_i - (a + bx_i)) x_i = 0$
	$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$